A Phonotactic Atlas of the World's Languages

Can we map languages by how their words *sound*?

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Every language has unwritten rules about how sounds may be strung together. English words can begin with *str-* but never with *tl-*; Japanese words almost always end in a vowel or *n*; Georgian happily starts words with clusters like *mtsv-*. These rules are called **phonotactics**. Every child learns them, but for most of the world's ∼7,000 languages nobody has written them down as statistics.

In this project you will measure phonotactics *from data* for as many languages as you can, turn each language into a numerical "sound profile", and use those profiles to draw a map of the world's languages built from sound alone. Then you will ask what that map actually shows.

Research Question

> When two languages "sound alike" phonotactically, is it mostly because they are **related** (same family), because they are **neighbors** (geographic contact), or because of **something else**?

Language family trees and coordinates are publicly available (e.g. **Glottolog**, `https://glottolog.org`), so there is ground truth to check against, and an interesting answer either way.

Data

You will work from **text**: pronunciation dictionaries that already pair written words with their transcription in the International Phonetic Alphabet (IPA). No audio, no spelling-to-sound conversion needed.

- **WikiPron** (`https://github.com/CUNY-CL/wikipron`). Wiktionary pronunciations for 300+ languages, one TSV per language: a word, a tab, and its IPA already split into segments (`casa  k a s a`). Prefer the `broad` (phonemic) files. Many words per language; transcription habits vary by language.
- **ASJP** (`https://github.com/lexibank/asjp`). A 40-word basic vocabulary for thousands of language varieties in one uniform 41-symbol alphabet. Huge coverage; few words each.

Python helpers worth knowing: `panphon` (IPA segment → articulatory features), `umap-learn` or `scikit-learn` (2D embeddings), `plotly` (world maps).

Getting started

Start small: load two languages you know, count how often each sound appears, and check that the top 10 look right before scaling up. From there, turn each language into a "phonotactic profile", a set of probability distributions you can compare across languages. Some suggestions:

- **Unigrams:** how often each segment occurs.
- **Bigrams with boundaries:** pairs of neighbors with `#` marking word edges, so `# k a s a #` gives `#k, ka, as, sa, a#`.
- **Edges:** which segments start and end words.
- **Word shapes:** replace each segment by C or V and count patterns like `CVCV`, or count consonant-cluster lengths.

Design choice to think about: if you compare raw segments, languages with different sound inventories look different almost by definition. Mapping segments to coarser classes (stop, fricative, nasal, vowel, . . .) puts every language in a shared space. Try both.

Once you have profiles, how you compare them, map them, and test the research question is up to you.